On the Correctness of the Real Parameterization, the Modular Caveat and the Conditional Deduction under the Hypothesis of Mutual Scale Symmetry

Statement of the Problem

This text serves as an explanation for the current version of the paper and the corresponding `GlobalNormalization.v` file. Its goal is to carefully separate four levels of reasoning:

$$\text{the real level},$$

$$\text{the integer level},$$

$$\text{the modular level},$$

$$\text{the level of mutual scale symmetry}.$$

On the real level, the following parameterization is considered:

$$z = m^n + p^n, \qquad x = m^n - p^n,$$

where the parameters

$$m,\ p$$

are initially not required to be integers. They can be treated as real parameters.

On the integer level, an additional question arises: when is such a real reconstruction compatible with the requirement

$$x, y, z \in \mathbb{N},$$

and, if necessary, with the requirement

$$m, p \in \mathbb{Z}?$$

This is exactly where parity and exact power conditions appear.

On the modular level, a simple but important methodological caveat is separately stated: congruence modulo is not the same as equality in integers. Therefore, the existence of solutions to congruences modulo does not in itself constitute a counterexample to the integer formulation of Fermat's Last Theorem.

Finally, the central conditional step of the current version is formulated through two real normalizations of the same odd binomial core:

$$\text{core} = n\,l^n, \qquad \text{core} = 2^{n-1} q^n.$$

Here

$$l > 0, \qquad q > 0$$

are real residual scales. Comparing these two representations gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

An additional structural condition is the condition of mutual scale symmetry

$$l^n = q^n.$$

It is precisely from this condition, together with the previous relation, that we obtain

$$n = 2^{n-1}.$$

Next, Coq verifies the elementary final step:

$$n = 2^{n-1} \quad \Longrightarrow \quad n \in \{1, 2\}.$$

Therefore, under the condition of mutual scale symmetry, the exponent

$$n > 2$$

is excluded.

It is important to emphasize immediately: this scheme remains conditional. Coq does not prove Fermat's Last Theorem unconditionally. It verifies the links between the formalized premises and the final exponential-linear obstacle. The crucial mathematical point remains the status of the condition

$$l^n = q^n.$$

1. Correctness of the Equalities for x^n and z^n

The equalities

$$x^n = (m^n - p^n)^n$$

and

$$z^n = (m^n + p^n)^n$$

are correct provided that it is preliminarily set that

$$x = m^n - p^n,$$

$$z = m^n + p^n.$$

This is the standard raising of equal quantities to the same power.

In other words, if

$$x = m^n - p^n,$$

then automatically

$$x^n = (m^n - p^n)^n.$$

Similarly, if

$$z = m^n + p^n,$$

then automatically

$$z^n = (m^n + p^n)^n.$$

It is not required here that

$$m, \qquad p$$

be integers. It is sufficient that the expressions are defined in the chosen numerical domain, for example, in the domain of real numbers.

However, if it is assumed in advance that

$$x, z \in \mathbb{N},$$

then the equalities

$$x = m^n - p^n, \qquad z = m^n + p^n$$

become compatibility conditions between the real parameterization and the natural values of

$$x, \qquad z.$$

In the real case, there might be no arithmetic restrictions, but in the integer case, additional parity and exact power conditions arise.

In Coq, this part is reflected by the lemma

`sum_diff_from_parameters_R`,

which establishes the real identities: if

$$z = m^n + p^n, \qquad x = m^n - p^n,$$

then

$$z + x = 2m^n, \qquad z - x = 2p^n.$$

On the natural number level, the lemma

`parameter_equations_raise_nat`

verifies an elementary fact: if

$$z = m^n + p^n, \qquad x = m^n - p^n,$$

are given, then after raising to the power, we automatically obtain

$$z^n = (m^n + p^n)^n, \qquad x^n = (m^n - p^n)^n.$$

Thus, there is no hidden mathematical assumption at this step: only the elementary property of equality is used.

2. Real Factorization of the Odd Binomial Core

Consider the difference

$$(m^n + p^n)^n - (m^n - p^n)^n.$$

By the binomial theorem, only the odd binomial terms remain in it:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2 \sum_{\substack{1 \le j \le n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Let us denote the inner sum by

$$S_n(m, p) = \sum_{\substack{1 \le j \le n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Then

$$y^n = 2S_n(m, p).$$

The first real normalization introduces a positive scale parameter

$$l$$

and writes the core in the form

$$S_n(m, p) = n\, l^n.$$

Equivalently, if

$$n > 0,$$

then

$$l^n = \frac{S_n(m, p)}{n}.$$

In this sense, the equality

$$y^n = 2n\, l^n$$

is correct as a real notation.

At the same time, l is not required to be an integer. If the positive real case is chosen, it is assumed that

$$l > 0.$$

In Coq, this is reflected by the definition

$$\texttt{real\_core\_factorization}\; n \;\text{core}\; l,$$

which means

$$0 < n, \qquad \text{core} = n\, l^n$$

over real numbers. This is not a statement about integer divisibility, but exactly a real factorization.

The lemma

```
real_core_factorization_substitutes_y_power
```

shows: if

$$y^n = 2\,\text{core}$$

and

$$\text{core} = n\, l^n,$$

then

$$y^n = 2n\, l^n.$$

This perfectly corresponds to the transition from the odd binomial core to the first scaled notation.

3. Distinction Between Real Factorization and Integer Divisibility

If

$$l$$

is treated as a real number, there is no need to assert that the sum

$$S_n(m,p) = \sum_{\substack{1 \le j \le n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j$$

is divisible by

$$n$$

in integers.

One should strictly distinguish between two statements:

$$l^n = \frac{S_n(m,p)}{n}$$

as a real definition, and

$$n \mid S_n(m,p)$$

as an integer arithmetic statement.

The first is permissible in the real model. The second requires a separate proof and is not true in the general case.

For example, for

$$n = 6, \qquad m = 1, \qquad p = 1$$

the odd sum of binomial coefficients is equal to

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32.$$

But

$$32$$

is not divisible by

$$6$$

in integers.

This does not destroy the real scheme, because it does not assert the divisibility of the core by

$$n$$

in the ring

$$\mathbb{Z}.$$

The real scheme uses the notation

$$S_n(m,p) = n\, l^n$$

over

$$\mathbb{R}.$$

This is exactly why the definition

```
real_core_factorization
```

speaks of representing the core in the form

$$\text{core} = n\, l^n$$

over real numbers, rather than

$$n$$

being an integer divisor of the core.

4. Integer Specialization and Parity Conditions

If it is additionally required that

$$m, p \in \mathbb{Z},$$

then from

$$z = m^n + p^n, \qquad x = m^n - p^n$$

it follows that

$$z + x = 2m^n, \qquad z - x = 2p^n.$$

Consequently,

$$z + x$$

and

$$z - x$$

must be even.

In the Coq language, this is expressed by the lemma

`sum_diff_from_parameters_Z`

and the corollary

`parity_condition_Z`.

If at least one of the parity conditions is violated, then integer parameters

$$m, p \in \mathbb{Z}$$

in this form are impossible. This is reflected by the lemmas

`no_parameters_if_parity_violation`, `no_parameters_if_odd`.

For instance, Coq verifies the specific case

$$z = 2, \qquad x = 1, \qquad n = 3.$$

Here

$$z + x = 3$$

is odd, so there are no integers

$$m, p$$

such that

$$2 = m^3 + p^3, \qquad 1 = m^3 - p^3.$$

This is formalized by the lemma

`no_parameters_for_example`.

Thus, Coq separates the general real level from the integer specialization. The real reconstruction may be formally correct, but the integer reconstruction requires additional arithmetic conditions.

5. Two Real Normalizations of the Odd Binomial Core

The central object of the current scheme is the odd binomial core

$$S_n(m,p) = \sum_{\substack{1 \le j \le n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

It arises from the equality

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m,p).$$

The first normalization isolates the linear binomial factor

$$n$$

and writes

$$S_n(m,p) = n\, l^n.$$

In Coq, this corresponds to

$$\texttt{real\_core\_factorization}\; n \;\text{core}\; l.$$

The second normalization isolates the total mass of the odd binomial coefficients

$$2^{n-1}.$$

The classical identity has the form

$$\sum_{\substack{1 \le j \le n \\ j \text{ odd}}} \binom{n}{j} = 2^{n-1}.$$

Therefore, one can introduce a second positive real residual scale

$$q$$

and write

$$S_n(m,p) = 2^{n-1} q^n.$$

Here

$$q$$

is a real scale parameter. It should not be confused with the modulus in modular arithmetic.

In Coq, the second normalization corresponds to the definition

$$\texttt{coefficient\_mass\_factorization}\; n \;\text{core}\; q,$$

which means

$$0 < n, \qquad \text{core} = 2^{n-1} q^n$$

over real numbers.

If the same core is written in two ways,

$$S_n(m,p) = n\, l^n, \qquad S_n(m,p) = 2^{n-1} q^n,$$

then

$$n\,l^n = 2^{n-1}q^n.$$

After dividing by

$$n > 0$$

we obtain the exact relation

$$l^n = q^n\frac{2^{n-1}}{n}.$$

In Coq, this is verified by the lemma

`two_scale_factorizations_relation`.

Its meaning is as follows: from two real factorizations of the same core, it automatically follows that

$$l^n = q^n\frac{2^{n-1}}{n}.$$

6. The Condition of Mutual Scale Symmetry

The central structural condition of the current version is the condition of mutual scale symmetry

$$l^n = q^n.$$

In Coq, it is defined as

`mutual_scale_symmetry` $n\ l\ q$.

This means exactly

$$l^n = q^n.$$

Comparing the two normalizations gives

$$l^n = q^n\frac{2^{n-1}}{n}.$$

If it is additionally satisfied that

$$l^n = q^n,$$

we get

$$q^n = q^n\frac{2^{n-1}}{n}.$$

Since

$$q > 0,$$

we have

$$q^n > 0.$$

Consequently, canceling by

$$q^n$$

is justified, leaving

$$1 = \frac{2^{n-1}}{n}.$$

Hence

$$n = 2^{n-1}.$$

In Coq, this transition is verified by the lemma

`mutual_scale_symmetry_forces_shift`.

It asserts: if the two factorizations

$$\mathrm{core} = n\, l^n, \qquad \mathrm{core} = 2^{n-1} q^n,$$

are satisfied, if

$$l^n = q^n$$

is satisfied, and if

$$q > 0,$$

then

$$n = 2^{n-1}.$$

For convenience, Coq also introduces the record

`MutualScaleData`.

This structure packs the data:

$$n, \quad \mathrm{core}, \quad l, \quad q,$$

the positivity

$$l > 0, \qquad q > 0,$$

the two factorizations

$$\mathrm{core} = n\, l^n, \qquad \mathrm{core} = 2^{n-1} q^n,$$

and the mutual scale symmetry condition

$$l^n = q^n.$$

The lemma

`mutual_scale_data_forces_shift`

deduces from this structure that

$$n = 2^{n-1}.$$

7. Elementary Exponential-Linear Obstacle

It remains to analyze the equation

$$n = 2^{n-1}.$$

It is equivalent to

$$2^n = 2n.$$

In positive natural numbers, this equation has exactly two solutions:

$$n = 1, \qquad n = 2.$$

Indeed,

$$1 = 2^{1-1} = 1,$$

$$2 = 2^{2-1} = 2.$$

And for any integer

$$n > 2$$

the exponential term

$$2^{n-1}$$

is strictly greater than the linear term

$$n.$$

In Coq, the corresponding part is broken down into several elementary lemmas:

`pow2_gt_linear_shift`,

`pow2_gt_linear`,

`pow_eq_linear_positive`,

`two_pow_eq_two_mul_iff_shift`,

`binary_scaling_roots_only_one_two`.

The final lemma

`binary_scaling_roots_only_one_two`

asserts:

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2.$$

At the level of the packed structure

`MutualScaleData`

this yields the lemma

`mutual_scale_data_forces_small_exponent`,

which asserts:

$$\texttt{MutualScaleData} \implies n = 1 \text{ or } n = 2.$$

Finally, the theorem

`mutual_scale_data_excludes_high_exponent`

states:

$$\texttt{MutualScaleData} \implies n > 2 \implies \text{False}.$$

That is, given the two factorizations and the mutual scale symmetry condition, an exponent higher than a square is excluded.

8. The Modular Caveat

The present scheme is not a proof via modular arithmetic and does not assert the absence of solutions to the corresponding congruences over finite fields.

The scheme under consideration relates to equalities over

$$\mathbb{N}, \qquad \mathbb{Z}, \qquad \mathbb{Q}, \qquad \mathbb{R},$$

rather than to equations over a finite field

$$\mathbb{F}_{\mathrm{modq}}.$$

The integer equality

$$x^n + y^n = z^n$$

indeed implies a congruence modulo any modulus

$$\mathrm{modq}:$$

$$x^n + y^n \equiv z^n \pmod{\mathrm{modq}}.$$

However, the reverse transition is generally false. The congruence

$$x^n + y^n \equiv z^n \pmod{\mathrm{modq}}$$

does not imply the equality

$$x^n + y^n = z^n$$

in integers.

In other words, a modular congruence only means that there exists an integer

$$t$$

such that

$$z^n = x^n + y^n + \mathrm{modq} \cdot t.$$

An integer solution to Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions to congruences

$$x^n + y^n \equiv z^n \pmod{\mathrm{modq}}$$

for

$$n > 2$$

is not a counterexample to Fermat's Last Theorem and does not refute the real parameterization used in this scheme.

If one specifically moves to the modular level, the parameters should be defined separately as residues. For example, instead of the real equalities

$$z = m^n + p^n, \qquad x = m^n - p^n$$

one must separately fix

$$m^n \equiv A \pmod{\mathrm{modq}}, \qquad p^n \equiv B \pmod{\mathrm{modq}}.$$

After that, the corresponding modular relations take the form

$$z \equiv A + B \pmod{\mathrm{modq}}, \qquad x \equiv A - B \pmod{\mathrm{modq}}.$$

Thus, the real parameterization and modular arithmetic belong to different levels of description.

9. How the Modular Caveat is Reflected in Coq

There is a section

$$\texttt{ModularRemark}$$

in Coq. In it, the relation of modular congruence is defined as

$$\texttt{modular\_congruent}\ \text{modq}\ a\ b.$$

Mathematically, this means that there exists an integer

$$t$$

such that

$$a = b + \text{modq} \cdot t.$$

The lemma

$$\texttt{integer\_equality\_implies\_modular\_power\_congruence}$$

verifies the direction

$$\text{integer equality} \quad \Longrightarrow \quad \text{modular congruence}.$$

If

$$x^n + y^n = z^n,$$

then one can take

$$t = 0,$$

and obtain the corresponding congruence modulo any

$$\text{modq}.$$

The lemma

$$\texttt{integer\_equality\_is\_zero\_modular\_witness}$$

explicitly establishes that a true equality in integers corresponds to a zero modular witness

$$t = 0.$$

The lemma

$$\texttt{modular\_congruence\_not\_integer\_equality}$$

verifies a specific example:

$$3^3 \equiv 1^3 + 1^3 \pmod 5,$$

since

$$27 = 2 + 5 \cdot 5,$$

but at the same time

$$1^3 + 1^3 \neq 3^3$$

as an equality in integers, since

$$2 \neq 27.$$

The structure

`ModularParameterResidues`

formalizes the idea that on the modular level one needs to separately define the residues

$$m^n \equiv A \pmod{\text{modq}}, \qquad p^n \equiv B \pmod{\text{modq}},$$

and then separately define

$$z \equiv A + B \pmod{\text{modq}}, \qquad x \equiv A - B \pmod{\text{modq}}.$$

Thus, Coq strictly confirms the methodological idea:

$$\text{integer equality} \Longrightarrow \text{modular congruence},$$

but

$$\text{modular congruence} \not\Longrightarrow \text{integer equality}.$$

10. General Structure of the Coq Verification

The Coq file verifies the conditional chain of the current version. It can be written as follows:

$$\text{core} = n\, l^n,$$

$$\text{core} = 2^{n-1} q^n,$$

$$l^n = q^n,$$

$$q > 0$$

imply

$$n = 2^{n-1}.$$

Then, elementary arithmetic of natural numbers yields

$$n = 2^{n-1} \quad \Longrightarrow \quad n \in \{1, 2\}.$$

In other words, Coq does not verify a single historical reconstruction in its entirety, but rather the formal correctness of the following bridges:

$$\text{two real factorizations} \Longrightarrow \text{exact ratio of scales},$$

$$\text{exact ratio of scales} + \text{mutual symmetry} \Longrightarrow n = 2^{n-1},$$

$$n = 2^{n-1} \Longrightarrow n = 1 \text{ or } n = 2.$$

The main Coq objects have the following meaning:

- `real_core_factorization` means

  $$\text{core} = n\, l^n$$

  over real numbers.

- `coefficient_mass_factorization` means

  $$\text{core} = 2^{n-1} q^n$$

  over real numbers.

- `mutual_scale_symmetry` means

$$l^n = q^n.$$

- `two_scale_factorizations_relation` deduces from the two factorizations

$$l^n = q^n \frac{2^{n-1}}{n}.$$

- `mutual_scale_symmetry_forces_shift` deduces from the two factorizations, the condition

$$l^n = q^n$$

and the positivity

$$q > 0$$

the equality

$$n = 2^{n-1}.$$

- `MutualScaleData` packs

$$n, \quad \text{core}, \quad l, \quad q,$$

the positivity

$$l > 0, \qquad q > 0,$$

the two factorizations, and the mutual scale symmetry condition.

- `mutual_scale_data_forces_shift` deduces from `MutualScaleData`

$$n = 2^{n-1}.$$

- `binary_scaling_roots_only_one_two` verifies

$$n = 2^{n-1} \Longrightarrow n = 1 \text{ or } n = 2.$$

- `mutual_scale_data_forces_small_exponent` deduces from `MutualScaleData`

$$n = 1 \text{ or } n = 2.$$

- `mutual_scale_data_excludes_high_exponent` deduces from `MutualScaleData`

$$n > 2 \Longrightarrow \text{False}.$$

This structure is fully consistent with the conditional form of the current paper: the crucial point is not the arithmetic of the equation

$$n = 2^{n-1},$$

but the mathematical status of the condition

$$l^n = q^n.$$

11. What Coq Does Not Prove

For the sake of honesty, it should be explicitly stated that the Coq file does not prove Fermat's Last Theorem unconditionally.

It does not automatically prove that any hypothetical solution to the equation

$$x^n + y^n = z^n$$

for

$$n > 2$$

necessarily implies the condition

$$l^n = q^n.$$

This condition is the central structural premise of the current reconstruction.

Coq verifies the following:

$$\text{core} = n\,l^n, \qquad \text{core} = 2^{n-1}q^n, \qquad l^n = q^n$$

imply

$$n = 2^{n-1},$$

and then

$$n \in \{1, 2\}.$$

Therefore, given these premises, the exponent

$$n > 2$$

is impossible.

Thus, the weak point of the scheme lies neither in the real parameterization, nor in the notation

$$\text{core} = n\,l^n,$$

nor in the modular caveat, nor in the Coq-verified final exponential-linear part. The crucial mathematical point remains the condition of mutual scale symmetry

$$l^n = q^n.$$

12. Alignment with the Current Paper

In the current paper, the main conditional result is formulated as follows: if a hypothetical positive natural solution to Fermat's equation

$$x^n + y^n = z^n, \qquad n > 2$$

admits the symmetric real parameterization under consideration, and if two natural normalizations of the same odd binomial core satisfy the condition

$$l^n = q^n,$$

then the equality

$$n = 2^{n-1}$$

arises. But this equality is possible only for

$$n = 1 \quad \text{and} \quad n = 2.$$

Therefore, the exponent

$$n > 2$$

is excluded under the specified conditional premise.

In this sense, the Coq file corresponds to the paper: it verifies exactly the final formal chain

$$\text{core} = n\, l^n,$$

$$\text{core} = 2^{n-1} q^n,$$

$$l^n = q^n,$$

$$n = 2^{n-1},$$

$$n \in \{1, 2\}.$$

At the same time, the modular caveat remains a separate methodological block: it does not participate in the proof of the final exponential-linear obstacle, but protects the reasoning from conflating different levels — integer equalities and modular congruences.

Conditional Theorem. Let, for some natural exponent

$$n > 0$$

the same odd binomial core admit two real factorizations

$$\text{core} = n\, l^n, \qquad \text{core} = 2^{n-1} q^n,$$

where

$$l > 0, \qquad q > 0.$$

If the condition of mutual scale symmetry

$$l^n = q^n,$$

is satisfied, then

$$n = 2^{n-1}.$$

Consequently,

$$n = 1 \quad \text{or} \quad n = 2.$$

In particular, under these premises, the case

$$n > 2$$

is impossible.

Final Conclusion. Understanding that

$$m, p, l, q \in \mathbb{R}$$

the equalities

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

as well as the real factorizations

$$\text{core} = n\, l^n, \qquad \text{core} = 2^{n-1} q^n$$

are correct as a formal real scheme.

The Coq file consistently formalizes this scheme, separately establishing the real factorization, the integer specialization, the modular caveat, and the final exponential-linear comparison.

The main verified chain has the form

$$\text{core} = n\, l^n, \qquad \text{core} = 2^{n-1} q^n, \qquad l^n = q^n$$

$$\Longrightarrow$$

$$n = 2^{n-1}$$

$$\Longrightarrow$$

$$n \in \{1, 2\}.$$

Consequently, under the condition of mutual scale symmetry, exponents

$$n > 2$$

are excluded.

Modular congruences for

$$n > 2$$

may exist, but they belong to a different level of the problem and are not integer solutions to Fermat's equation.

Thus, the final status of the current scheme is clear: the final Coq-verified part is correct, and the decisive mathematical question is centered on justifying the structural condition

$$l^n = q^n.$$